

NCIE-004 v1.1 → v1.2

Amendment Verification Report

VPF & Dynamic Agent Synthesis Engineering Consistency Amendment — IN DEVELOPMENT / FOR HUMAN REVIEW

1. Scope Verification

All 36 chapters are present, in original order, confirmed by a programmatic heading scan (regex match on 'NCIE-004 — Chapter N Expansion' across the final document, N = 1..36, each found exactly once). No chapter was added, removed, duplicated or renumbered. Substantive amendments are confined to Chapters 1, 4, 18, 19, 20, 23, 24, 25, 26, 27, 28, 29, 30, 33, 34, 35 and 36 — the locked scope of the Consolidated Claude Production Directive v2 §2 — plus front matter and the running header (§15, below).

2. 36-Chapter and Structural Verification

Chapter headings, section-numbering convention (new subsections inserted as the next available decimal, e.g. Ch.20 gains §20.4-§20.9 following its existing §20.1-§20.3; no existing section was renumbered) and the overall Table/Figure sequence were verified intact. Word's TOC field was updated via the application (not hand-edited) so navigation reflects the amended structure; page count moved from the v1.1 baseline to 82 pages as a direct, expected consequence of the inserted content, not a formatting change.

3. Substantive Diff Verification

A table-row-level diff confirms 31 new Technology Decision rows (Ch.4: 3, Ch.18: 2, Ch.19: 3, Ch.20: 8, Ch.23: 2, Ch.24: 1, Ch.25: 2, Ch.26: 2, Ch.27: 1, Ch.28: 2, Ch.29: 3, Ch.30: 2), 1 new Technology Registry candidate (Ch.33: Open Policy Agent), 5 new deployment-topology relationship rows (Ch.34), 2 existing rows edited in place with preserved history (TD-20-3, TD-20-4), and 16 new prose subsections (Ch.1 §1.4; Ch.4 §4.3; Ch.18 §18.4; Ch.19 §19.3; Ch.20 §20.4-§20.9 [6 subsections]; Ch.23 §23.2; Ch.27 §27.2; Ch.33 §33.4; Ch.34 §34.4; Ch.35 §35.6; Ch.36 §36.2A, §36.5). No row or paragraph outside these locations was altered.

4. Agent Factory Verification

Ch.20 §20.4 and TD-20-6 define Agent Factory as a logical governed capability (Definition construction/compilation, structural validation, policy compilation, Gate A coordination, sandbox/evaluation coordination, activation preparation, artifact integrity) explicitly not assumed to be one monolithic service. Capability discovery precedes novel synthesis; a missing primitive routes to an Engineering Change Candidate rather than being fabricated. Codex re-entry boundary preserved verbatim: routine composition of approved primitives requires no bespoke Codex coding per Agent.

5. Ephemeral Runtime Verification

Ch.4 (TD-4-6/7/8, §4.3) and Ch.20 §20.5 (TD-20-7) define the Ephemeral Agent Execution Plane as bounded, technically enforced (identity, authorization, context, model, tool, network, resources, persistence, lifetime), with no administrative write authority over any Registry or Agent Factory governance. AGENT RUNTIME IDENTITY ≠ HUMAN IDENTITY and EPHEMERAL RUNTIME EXPIRY ≠ AUDIT/PROVENANCE DELETION are both stated verbatim.

6. VPF Gate A Verification

Ch.20 §20.6 and TD-20-9 state Gate A validates the applicable synthesized-Agent capability envelope before novel activation and does not register the Agent, approve future output, or create institutional authority — GATE A VALIDATES AGENT ENVELOPE — NOT FUTURE OUTPUT is preserved verbatim. Gate A and Sandbox/Evaluation are kept as explicitly separate controls (§20.6).

7. VPF Gate B Verification

Ch.20 §20.6 and TD-20-10 state Gate B validates applicable substantive AI-generated output before release as a Validated NCIE AI Result, uniformly across ARGUS direct use, Registered Agents, Ephemeral Agents and multi-Agent composite output — no agent type is represented as exempt. GATE B VALIDATION ≠ HUMAN APPROVAL is preserved verbatim, and COMPONENT PASS ≠ COMPOSITE PASS is stated for multi-Agent output (§20.8).

8. Model Gateway Verification

Ch.18 TD-18-7/8 and §18.4 confirm AGENT → MODEL GATEWAY → ELIGIBLE MODEL DEPLOYMENT, never AGENT → ARBITRARY PROVIDER ENDPOINT, reusing the existing §18.2 fallback/outage policy rather than creating a parallel one. Route provenance sufficient for NCIE-003 v0.4 (model/deployment ID, version, provider/location, eligibility context, invocation time, correlation reference) is required.

9. Tool Gateway Verification

Ch.20's pre-existing Tool Registry, Tool Gateway enforcement, Tool Gateway Enforcement Checklist (§20.3, 9 items) and Execution Receipts rows (TD-20-1, TD-20-2, TD-20-5) are unchanged. The new Agent Factory/Execution Plane content explicitly reuses this architecture (§20.7: 'AGENT → TOOL GATEWAY → AUTHORIZED TOOL' is not restated as a new rule but the existing checklist is cited by reference) — no duplicate or competing Tool Gateway model was introduced.

10. Registry Verification

Ch.20 §20.7 and TD-20-11/12 define the Agent Registry, Agent Pattern Registry and VPF Profile Registry with SYNTHESIZED ≠ REGISTERED and SHARED PHYSICAL TECHNOLOGY ≠ SHARED LIFECYCLE / APPROVAL AUTHORITY preserved verbatim. Ch.33 §33.4 confirms these reuse the existing PostgreSQL/backend-runtime stack rather than introducing new products, with Open Policy Agent (§33.4, Table 32) the sole new Registry candidate — reviewed and justified as genuinely cross-cutting (RBAC/ABAC/PBAC and VPF deterministic validation both depend on it) rather than assumed by default.

11. IAM / Secrets / Network Verification

Ch.23 TD-23-7/8 and §23.2 add Agent Runtime Identity as a distinct class (DELEGATED TASK ≠ HUMAN CREDENTIAL DELEGATION; REMEMBERED AUTHORIZATION ≠ CURRENT AUTHORIZATION) without redesigning IAM. Ch.24 TD-24-6 extends secret scoping to Agent Factory/Runtime/VPF. Ch.25 TD-25-7/8 add Agent egress deny-by-default and prompt-injection/instruction-authority containment (EVIDENCE ≠ INSTRUCTION; TOOL OUTPUT ≠ POLICY; AGENT OUTPUT ≠ AUTHORITY).

12. Observability / Audit Verification

Ch.26 TD-26-7/8 add Agent Factory/Runtime/VPF as a sixth non-collapsed telemetry signal category alongside the existing five, plus a telemetry-privacy rule. Ch.27 TD-27-6 and §27.2 route material Agent/VPF lifecycle events into the existing authoritative audit architecture (NCIE-003 v0.4 Ch.27) rather than a separate Agent-only event store; TELEMETRY ≠ AUDIT ≠ PROVENANCE and SECURITY EVENT ≠ REGULATORY / INSTITUTIONAL FINDING are both preserved.

13. CI/CD Verification

Ch.28 TD-28-8/9 extend the existing CI/CD/environment-tier model to Agent Patterns, VPF Profiles and Runtime Policies as governed versioned artifacts (DEPLOYED CONFIGURATION ≠ GOVERNANCE APPROVAL) with rollback support (SOFTWARE ROLLBACK ≠ GOVERNANCE HISTORY ROLLBACK), reusing the existing policy-as-code gate rather than creating a new promotion mechanism.

14. Testing Verification

Ch.29 TD-29-9/10/11 add the mandatory positive Dynamic Agent acceptance scenario (capability discovery → synthesis → Gate A → sandbox → activation → Gate B → expiry, no bespoke Codex coding), adversarial/negative tests (permission expansion, self-registration, recursive synthesis, Tool/VPF Gateway bypass) including the mandatory Human-Primary adversarial scenario, and fail-closed/reconciliation tests (mandatory-validator-unavailable, Tool-outcome-unknown, VPF historical-state-survives-revalidation).

15. Recovery Verification

Ch.30 TD-30-7/8 extend backup/HA/DR to Agent Factory, both Registries, Ephemeral Runtime, VPF Runtime and Validation Artifacts, and add explicit post-restore reconciliation of authorization/eligibility before dependent execution resumes. RECOVERY ≠ REAUTHORIZATION is preserved verbatim — restore never automatically reactivates expired/suspended Agents, retired Tools, ineligible Models or obsolete VPF Profiles.

16. Technology Registry Verification

Ch.33's Table 32 grows from 16 to 17 entries (Open Policy Agent added, Security/Governance category, Proposed — pending approval; no entry promoted to Approved by this amendment, consistent with §33.3's promotion invariant). §33.4 documents the reuse-first review that preceded this single addition. Ch.36 §36.3's snapshot count was updated to match (17 seeded technologies).

17. Technology Decision Matrix Reconciliation

Table 33 (summary) was recomputed programmatically from the final Table 34 (Decision Matrix) and Table 35 (Human Review Register) cell content — not hand-counted, and not carried forward from v1.1. Result: 196 total decisions (was 165) = 33 Approved Upstream Mandate + 7 new mandates = 40; 115 Proposed Design Default + 22 new (net of TD-20-3's reclassification) = 136; Institutional Confirmation Required unchanged at 6; Infrastructure Discovery Required 5 + 2 new = 7; Security/Sovereignty Confirmation Required unchanged at 6; Superseded (new category) = 1 (TD-20-3). Arithmetic reconciles: 165 + 31 added – 0 removed = 196, with TD-20-3 moving from the Proposed Design Default bucket into a new

Superseded bucket (a reclassification, not a net addition or removal). Blocking count 17 → 19 (+2: TD-4-8 and TD-20-13, both Infrastructure Discovery Required); Non-Blocking 115 → 137 (+22).

18. Human Review Reconciliation

The Human Review Register (Table 35) holds every non-mandate decision: 132 (v1.1) → 156 (v1.2) = 132 + 24 added (of the 31 new decisions, 7 are Approved Upstream Mandate and therefore do not enter this register) – 0 removed, reconciling exactly. TD-20-3 and TD-20-4's Human Review Register rows were updated in place (capability text annotated Superseded/Clarified) rather than duplicated.

19. Deployment Topology Verification

Ch.34's Figure 7 gains 5 relationship rows placing Agent Factory, the Ephemeral Execution Plane and VPF Gate B into the existing zone model without introducing a new zone. §34.4's diagram audit confirms — and this report independently re-confirms by inspecting the final row text — that no row states or implies Agent → arbitrary model provider, Agent → direct consequential external system, Agent → unrestricted Institutional Memory, Agent → Registry administrative write, Agent → self-registration, Agent → recursive synthesis, or VPF → institutional approval; the added rows explicitly state the negative form of each prohibited path (e.g. 'reaches models/tools only via Model Gateway / Tool Gateway (never directly)').

20. Generated-Code Status Verification

Every reference to generated-code execution across the amended document (Ch.20 §20.9, TD-20-13, Ch.35 Table 34/35, Ch.36 §36.2A/§36.5) reports one coherent disposition: NOT ACTIVATED BY NCIE-004 v1.2, classified INFRASTRUCTURE DISCOVERY REQUIRED / BLOCKING for generated-code activation specifically, and explicitly stated not to block configuration/composition-based Dynamic Agent Synthesis, whose independent controls (Ch.4, Ch.18-20, Ch.23-30) are satisfied. No occurrence contradicts this.

21. Traceability Verification

Ch.36's Table 36 traceability matrix rows for Ch.4, 18, 19, 20, 23-30, 33, 34 and 35 (15 rows) were each extended — not replaced — with their NCIE-007 v1.1 dependency (and, for Ch.20, the full NCIE-001/002/003/007 chain). Ch.1-3, 5-17, 21-22, 31-32 and 36 rows are unchanged. No orphan Agent/VPF requirement was found without a citation to at least one current approved authority.

22. Version & Document-Control Verification

Cover: 'Traces to approved NCIE-001 v1.3 ... approved NCIE-002 v0.5 ... approved NCIE-003 v0.4 ... approved NCIE-007 v1.1' — confirmed present, no v1.1/v0.4/v0.3/v1.2-as-current-NCIE-001 reference remains on the cover. Document Governance table: Edition, Requirements authority, Architecture authority, Canonical data authority, Orchestration authority (new), Prior baseline (new) and Status all confirmed showing v1.2/v1.3/v0.5/v0.4/v1.1 as applicable — confirmed programmatically by reading the final table cells, not by visual sampling. Filename: NCIE-004_Technology_Stack_Engineering_Blueprint_v1_2_IN_DEVELOPMENT.docx — confirmed. Supersedes / Prior baseline: correctly identifies the actual full NCIE-004 v1.1 (IN DEVELOPMENT — FOR HUMAN REVIEW) document, not an earlier skeletal edition. Status: IN DEVELOPMENT — FOR HUMAN REVIEW OF SURGICAL AMENDMENT (not Approved) — confirmed.

23. Running Header/Footer — Page-Level Version Scan

This is reported in full because the first-pass render of this amendment initially failed this exact check: after all substantive chapter amendments and front-matter/governance-table corrections were applied, the running header (a separate document part, word/header1.xml, untouched by any body-text edit) still read 'NCIE-004 | TECHNOLOGY STACK & ENGINEERING BLUEPRINT | VERSION 1.1' on every page. This was caught by rendering the document to PDF via Microsoft Word and text-scanning all 82 pages for the running header string before delivery — exactly the mandatory pre-delivery check the correction directive's document-control gate (§338-346, §407) exists to enforce. The header text run was corrected to VERSION 1.2 and the document was re-rendered; a second full page-level scan of all 82 pages found the string 'NCIE-004 | TECHNOLOGY STACK & ENGINEERING BLUEPRINT | VERSION 1.2' as the sole variant present, and zero occurrences of VERSION 1.1 in any running header. Required final stale count: 0 — confirmed.

A second, independent document-control defect was also found and corrected: the Document Governance table's Edition field read 'In development — Version 1.0' in the v1.1 baseline itself (the cover of that same v1.1 document already correctly said 'Version 1.1' — the two fields were already inconsistent before this amendment began). This is recorded in the Change Register as AM-DOC alongside the header fix, both classified DOCUMENT-CONTROL CORRECTION.

24. Full-Document Stale-Authority Scan

Every occurrence of 'v1.1' in the final document body and tables (37 occurrences) was individually inspected. All are either (a) the correctly-current citation to approved NCIE-007 v1.1 (NCIE-007 has not moved version), or (b) explicitly historical content: the pre-existing '0.6 Hardening Pass (v1.1)' heading and its body text, the new '0.7 Document Amendment History'

section's 'Version: v1.1' historical entry, the new 'Prior baseline' governance-table row, and TD-20-3's preserved historical position text (explicitly labelled 'HISTORICAL' in-line). No current-authority statement anywhere in the document cites NCIE-001 v1.2, NCIE-002 v0.4, NCIE-003 v0.3 or NCIE-004 v1.1 as a current reference.

25. Semantic Contradiction Scan

A full-text scan for statements implying Workflow Engine is the sole possible Agent runtime, Agent may self-register, recursive synthesis is normally allowed, VPF is optional for substantive AI output, VPF PASS equals Human Approval/Finding/Decision, Agent may bypass the Model or Tool Gateway, generated-code Agents are approved, or Agent Factory failure disables deterministic NCIE, found none in current-edition text. Every governing invariant listed in the directive's §3 (Ch.1 §1.4, Ch.20 §20.4-§20.9) is stated explicitly and preserved consistently wherever the corresponding capability is discussed.

26. Visual Verification

The complete amended document was rendered to PDF via Microsoft Word (82 pages) with Word's own field/TOC update applied first. Cover, Document Governance, Chapter 1, Chapter 18, Chapter 19, Chapter 20, the amended Ch.23-30 sections, Chapter 33, Chapter 34, Chapter 35 and Chapter 36 were confirmed present at their expected locations with no truncated table rows, no missing columns and no duplicate headings, verified via programmatic table/row/column-count checks on the final saved document (197 rows in Table 34 = 1 header + 196 data rows; 157 rows in Table 35 = 1 header + 156 data rows, both matching the reconciled totals exactly) rather than by visual sampling alone. No diagrams (image-based figures) were added or altered by this amendment — the new deployment-topology content (Ch.34) is expressed as table rows, consistent with this document's existing figure-as-table convention, so no image-clipping risk was introduced.

27. No Placeholder or Fabricated Content

A full-text scan for TODO, TBD, PLACEHOLDER, FIXME and unresolved template markers (including this build's own internal {TOTAL}/{NEW_COUNT} substitution tokens, confirmed resolved before delivery) found zero matches in the final document. No fabricated infrastructure, NFR values, institutional ownership, security/legal approval, implementation maturity or acceptance evidence was found for any new Agent/VPF requirement; INFRASTRUCTURE DISCOVERY REQUIRED and the existing four other controlled statuses are used consistently in their place — no competing 'ENGINEERING / INFRASTRUCTURE DISCOVERY REQUIRED' alias was left in the delivered document (an internal drafting inconsistency of that kind was found and normalized to the existing v1.1 status vocabulary before this build was finalized).

28. Three-Artifact Consistency Check

The amended NCIE-004 v1.2 main document, this Verification Report and the accompanying Surgical Amendment Change Register agree on: Source Version v1.1 → Target Version v1.2; Authority chain approved NCIE-001 v1.3 / NCIE-002 v0.5 / NCIE-003 v0.4 / NCIE-007 v1.1; Status IN DEVELOPMENT — FOR HUMAN REVIEW OF SURGICAL AMENDMENT (not Approved); 36 chapters preserved; amendment footprint = Chapters 1, 4, 18-20, 23-30, 33-36 plus front matter/running header; Technology Decision Matrix 165 → 196 (31 added, 0 removed, 1 reclassified); Human Review Register 132 → 156 (19 Blocking / 137 Non-Blocking); TD-20-3 = SUPERSEDED / AMENDED (historical position preserved); TD-20-4 = CLARIFIED (scope unchanged, not broadened); Generated-Code Status = NOT APPROVED BY NCIE-004 v1.2, does not block configuration/composition-based synthesis.

29. Disposition

NCIE-004 — Technology Stack & Engineering Blueprint v1.2 — VPF & Dynamic Agent Synthesis Engineering Consistency Amendment is READY FOR HUMAN REVIEW. The v1.1 approved-pending-review baseline remains historically preserved and in force except for the specific amendments listed in the accompanying Change Register. This document is NOT marked Approved.